STATISTICS AND PROBABILITY

CHAPTER 13

Probability Distributions

13

SYLLABUS 4.7 4.8 4.9 4.12 4.14

Measure the same kind of thing many times and you rarely get the same answer twice. Masses differ from packet to packet, heights differ from person to person, and a repeated experiment gives a different count each time. One number cannot describe a quantity that behaves like this. What describes it is the list of values it can take together with how likely each one is, and that list is called a **distribution**. This chapter is about the few distributions that cover a great many situations.

A factory fills bags of rice, each meant to hold 500 grams, and weighs 5000 of them. No two masses agree exactly. Grouping the readings and drawing them as a graph gives a shape: most bags near 500 g, with fewer and fewer further out on either side, falling away symmetrically into one smooth hump. A bell.

The same curve turns up in places that have nothing to do with rice. Astronomers found it in the *errors* of their telescope readings, and biologists in the heights of people. The reason is the same in every case. A bag's mass is not decided by one thing but by many: the size of the individual grains, a slight vibration in the machine, the exact instant the valve closes. Each effect is tiny and as likely to push the mass up as down, and when many such effects add together, the bell is what you get.

A distribution is worth having because it turns uncertainty into something you can calculate with. From one you can predict an average result, measure how far individual results are likely to stray from it, and put a number on the risk of an extreme value. That is what makes these few curves useful in medicine, manufacturing and finance. It is also where care is needed, because every answer depends on the distribution being the right description of the situation, and choosing the wrong one gives confident answers that are wrong.

We start with the simplest case, a single uncertain number taking a few values, and work up through the distribution used for coin flips and quality control, until the bell curve appears as their limit.

13.1 Discrete Random Variables

A **random variable** is a number whose value depends on chance. The number of heads in three coin tosses, the score on a die, the number of defective items in a box: each is a rule that assigns a number to a random outcome. We write random variables with capital letters, X, and particular values with lower case, x.

A random variable is **discrete** when it can only take separate, countable values. To describe it fully, you list every value it can take alongside the probability of each. This list is its **probability distribution**.

DEF A **discrete random variable** X takes countable values $x_1, x_2, \ldots$ Its **probability distribution** gives $P(X = x_i)$ for each value, subject to

$$P(X = x_i) \geq 0 \qquad \text{and} \qquad \sum_i P(X = x_i) = 1.$$

The second condition is the one you use most often. Since some value must occur, all the probabilities add to exactly 1. This single fact lets you find a missing probability.

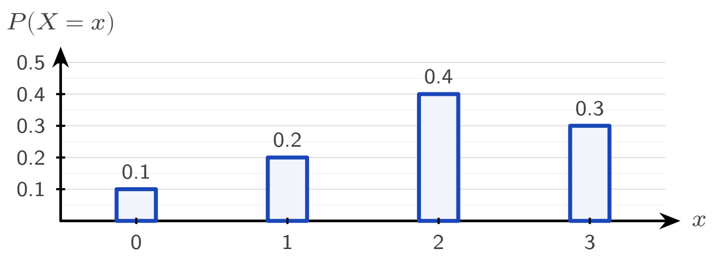

Figure 13.1 The probability distribution of a discrete random variable. Each bar is the probability of one value, and the four bars must total exactly 1.

EXAMPLE 13.1

A discrete random variable X has the distribution below. Find k, then $P(X \geq 3)$.

x	**1**	**2**	**3**	**4**
$P(X = x)$	0.1	0.3	k	0.2

The probabilities must sum to 1:

$$0.1 + 0.3 + k + 0.2 = 1 \implies k = 0.4.$$

Then $P(X \geq 3) = P(X = 3) + P(X = 4) = 0.4 + 0.2 = 0.6$.

EXAMPLE 13.2

A random variable X takes values $1, 2, 3, 4$ with $P(X = x) = kx$. Find k.

Sum the probabilities and set equal to 1:

$$k(1) + k(2) + k(3) + k(4) = 10k = 1 \implies k = \frac{1}{10}.$$

When the distribution is given by a formula, the total-probability condition still determines the constant.

YOUR TURN 13.1 Discrete Random Variables

1.

(a) For $x = 1, 2, 3$ the probabilities are 0.2, 0.5 and k. Find k. [1]

(b) A variable X has $P(X = x) = 0.15,\ 0.25,\ 0.35,\ k$ for $x = 0, 1, 2, 3$. Find k and $P(X \leq 1)$. [2]

(c) A variable takes values $x = 0, 1, 2, 3$ with $P(X = x) = k(x + 1)$. Find k. [2]

(d) A variable takes values $x = 1, 2, 3, 4$ with $P(X = x) = \dfrac{x^2}{30}$. Find $P(X = 3)$. [2]

(e) For the distribution $P(X = x) = 0.1,\ 0.4,\ 0.3,\ 0.2$ at $x = 0, 1, 2, 3$, find $P(X > 1)$ and $P(1 \leq X \leq 2)$. [2]

2.

(a) Show that $P(X = x) = \frac{1}{18}(4 + x)$ for $x = 1, 2, 3$ defines a probability distribution, and find $P(X \geq 2)$. [3]

(b) A variable takes values $x = 1, 2, 3, 4$ with probabilities 0.1, a, 0.3, b. Given that $P(X \leq 2) = 0.4$, find a and b. [3]

(c) Hence find $P(X > 2)$ for the distribution in part (b). [1]

(d) A variable takes values $x = 1, 2, 3$ with $P(X = x) = k \cdot 2^x$. Find k. [2]

13.2 Expected Value and Variance

If you played a game many times, what would you win on average per play? That long-run average is the **expected value** $E(X)$. It is not the most likely outcome, nor a value X need ever take. It is the balance point of the distribution: each value weighted by its probability.

KEY · IB

$$E(X) = \mu = \sum_i x_i \, P(X = x_i)$$

This is exactly the mean from Ch. 11, with probabilities playing the role of relative frequencies. Over many trials, the relative frequency of each value approaches its probability, so the average approaches $E(X)$.

EXAMPLE 13.3

A game costs \$2 to play. You roll a fair die and win, in dollars, the number shown. Is the game worth playing?

Let X be the amount won. Each face is equally likely:

$$E(X) = \tfrac{1}{6}(1+2+3+4+5+6) = \tfrac{21}{6} = 3.5.$$

You expect to win \$3.50 per play but pay \$2, so the expected net gain is \$1.50. Over many games you gain money on average.

That last calculation names an idea the syllabus uses directly. A game is **fair** when the expected net gain is zero, so that neither side profits in the long run. Since the net gain is the winnings minus the stake, a fair stake is exactly the expected winnings.

KEY · IB A game is **fair** when $E(\text{gain}) = 0$, that is, when the stake equals $E(\text{winnings})$.

EXAMPLE 13.4

A fair die is rolled. You win \$10 for a six, \$4 for a four or a five, and nothing otherwise. Find the stake that makes the game fair, and state whether a stake of \$4 favours the player or the organiser.

Let W be the amount won. The three cases have probabilities $\frac{1}{6}$, $\frac{2}{6}$ and $\frac{3}{6}$, so

$$E(W) = 10\left(\tfrac{1}{6}\right) + 4\left(\tfrac{2}{6}\right) + 0\left(\tfrac{3}{6}\right) = \tfrac{10+8}{6} = 3.$$

A stake of \$3 makes the game fair: the expected gain is $3 - 3 = 0$. At a stake of \$4 the expected gain is $3 - 4 = -1$, a loss of one dollar per play on average, so the game favours the organiser. Note that no single play ever produces a loss of exactly \$1; the figure is the average over many plays.

Variance and standard deviation HL

Expected value locates the centre; **variance** measures the spread, just as in descriptive statistics. The variance is the expected squared distance from the mean. A short calculation turns it into a more practical form.

KEY · IB HL

$$\text{Var}(X) = E((X-\mu)^2) = E(X^2) - \mu^2, \qquad \text{where } E(X^2) = \sum_i x_i^2\, P(X = x_i)$$

The form $E(X^2) - \mu^2$ is mostly easier to use: find the mean of the squares, then subtract the square of the mean. The standard deviation is $\sigma = \sqrt{\text{Var}(X)}$.

EXAMPLE 13.5

Find the variance and standard deviation of X:

x	**0**	**1**	**2**	**3**
$P(X = x)$	0.1	0.2	0.4	0.3

First the mean:

$$\mu = 0(0.1) + 1(0.2) + 2(0.4) + 3(0.3) = 1.9.$$

Then $E(X^2)$:

$$E(X^2) = 0^2(0.1) + 1^2(0.2) + 2^2(0.4) + 3^2(0.3) = 0.2 + 1.6 + 2.7 = 4.5.$$

So $\text{Var}(X) = 4.5 - 1.9^2 = 4.5 - 3.61 = 0.89$, and $\sigma = \sqrt{0.89} \approx 0.943$.

Linear transformations HL

If you scale and shift a random variable, its mean and variance respond predictably, exactly as a data set does. For constants a and b:

KEY · IB HL

$$E(aX + b) = aE(X) + b, \qquad \text{Var}(aX + b) = a^2\, \text{Var}(X)$$

The shift b moves the mean but not the spread. The scale a stretches the mean and stretches the variance by a^2, since variance is built from squared distances.

EXAMPLE 13.6

For the variable above, $E(X) = 1.9$ and $\text{Var}(X) = 0.89$. Let $Y = 2X + 3$. Find $E(Y)$ and $\text{Var}(Y)$.

$$E(Y) = 2(1.9) + 3 = 6.8, \qquad \text{Var}(Y) = 2^2(0.89) = 3.56.$$

The $+3$ leaves the variance untouched; only the factor of 2, squared, scales it.

YOUR TURN 13.2 Expected Value and Variance

3.

(a) Find $E(X)$ for $x = 1, 2, 3$ with $P(X = x) = 0.2,\ 0.3,\ 0.5.$ [2]

(b) Find $E(X)$ for $x = 0, 1, 2, 3$ with $P(X = x) = 0.1,\ 0.4,\ 0.3,\ 0.2.$ [2]

(c) A fair coin pays $3 for heads and loses $2 for tails. Find the expected gain per toss. [2]

(d) A game costs $1. You win $5 with probability $\frac{1}{8}$, and otherwise lose your stake. Find the expected gain, and state whether the game is fair. [3]

4.

(a) For the distribution in question 3 part (b), find $E(X^2)$ and Var(X). [3]

(b) Given $E(X) = 4$ and Var$(X) = 2$, find $E(3X - 1)$ and Var$(3X - 1)$. [2]

5.

(a) A wheel is spun for a prize. It pays $8 with probability 0.1, $2 with probability 0.3, and nothing otherwise. Find the expected amount won, and the stake that would make the game fair. [3]

(b) The stall charges $2 to spin. Find the expected gain per spin for a player, and the expected loss over 50 spins. [3]

(c) A variable has $E(X) = 3$ and Var$(X) = 5$. Find $E(2X + 1)$, Var$(2X + 1)$ and $E(X^2)$. [4]

13.3 The Binomial Distribution

Many situations are the same experiment repeated: ten coin flips, twenty items checked for defects, a quiz of multiple-choice guesses. When each repetition is independent and has just two outcomes, the count of "successes" follows one specific distribution, the **binomial**.

Four conditions define it. Check each one before reaching for the distribution.

- **Fixed number of trials.** The value of n is decided in advance, not by the results.
- **Two outcomes only.** Every trial ends as a success or a failure, with nothing in between.
- **Constant probability.** Every trial has the same probability of success p.

- **Independence.** The result of one trial does not change the probabilities for any other.

When all four hold, write $X \sim B(n,p)$.

DEF If X counts the successes in n independent trials each with success probability p, then $X \sim B(n,p)$, and

$$P(X=r)=\binom{n}{r}p^r(1-p)^{n-r}, \qquad r=0,1,2,\dots,n.$$

Each factor has a meaning: p^r for r successes, $(1-p)^{n-r}$ for the rest failing, and $\binom{n}{r}$ for the number of different orders in which those successes can fall. In practice you evaluate it with a GDC, using the binomial pdf for $P(X=r)$ and the binomial cdf for $P(X \le r)$.

EXAMPLE 13.7

A student guesses on a 10-question multiple-choice test, each question having 4 options. Find the probability of (a) exactly 3 correct, (b) at most 2 correct.
Here $X \sim B(10, 0.25)$, since each guess is correct with probability $\frac{1}{4}$.
(a) Using the binomial pdf:

$$P(X=3)=\binom{10}{3}(0.25)^3(0.75)^7 \approx 0.250.$$

(b) Using the binomial cdf:

$$P(X \le 2)=P(0)+P(1)+P(2) \approx 0.526.$$

Despite 10 questions, scoring 3 or fewer is more likely than not. Guessing rarely produces a good score.

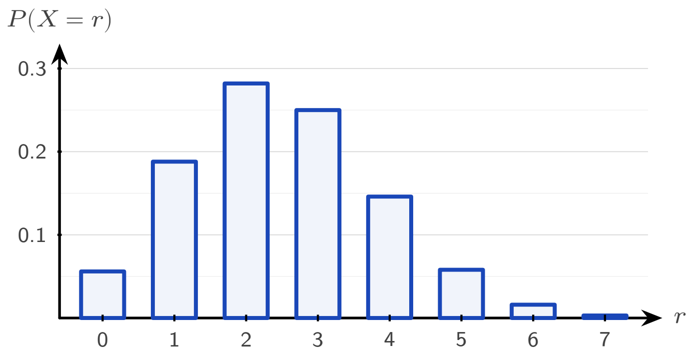

Figure 13.2 The distribution $B(10,\ 0.25)$, the guessing student of the example above. The bars are tallest at $r=2$ and $r=3$, and the long tail to the right shows why a high score by guessing is so unlikely.

Mean and variance of the binomial

You could find the mean by summing $r\,P(X=r)$, but there is no need. For n trials each succeeding with probability p, the results are clean.

KEY · IB For $X \sim B(n,p)$:

$$E(X) = np, \qquad \text{Var}(X) = np(1-p)$$

The mean is intuitive: 10 trials at probability 0.25 give 2.5 expected successes. The variance formula is largest when $p=0.5$, when each trial is most unpredictable.

EXAMPLE 13.8

A factory's items are defective with probability 0.05, independently. A sample of 20 is taken. Find the expected number of defectives and the probability that at least one is defective.

Let $X \sim B(20, 0.05)$. The expected number is

$$E(X) = 20 \times 0.05 = 1.$$

For "at least one," use the complement:

$$P(X \geq 1) = 1 - P(X=0) = 1 - (0.95)^{20} \approx 0.642.$$

Even with a low defect rate, a sample of 20 is more likely than not to contain a defective item. The probability of "at least one" rises quickly as the sample size increases.

CAUTION Check independence and a fixed n before using the binomial. Drawing items *without replacement* breaks independence, since each draw changes the next probability, so that count is not binomial.

YOUR TURN 13.3 The Binomial Distribution

6. GDC

(a) $X \sim B(8, 0.5)$. Find $P(X = 4)$. [2]
(b) $X \sim B(10, 0.3)$. Find $P(X = 2)$. [2]
(c) $X \sim B(6, 0.4)$. Find $P(X \leq 2)$. [2]
(d) $X \sim B(12, 0.25)$. Find $P(X \geq 1)$. [2]

7.

(a) $X \sim B(20, 0.1)$. Find the mean and the variance. [2]
(b) $X \sim B(15, 0.6)$. Find $E(X)$ and Var(X). [2]

8. GDC In a factory, 8% of items are defective, independently of one another. A sample of 25 items is taken.

(a) Find the probability that exactly 2 items are defective. [2]
(b) Find the probability that at least 1 item is defective. [2]
(c) Find the expected number of defective items in the sample. [1]
(d) A quiz has 12 questions, each with 5 options, and a student guesses every answer. Find the probability of at least 4 correct answers. [3]
(e) A box holds 10 items, 3 of them defective, and 4 items are drawn without replacement. Explain why the number of defective items drawn is not binomial. [2]

13.4 Continuous Random Variables HL

A discrete variable jumps between separate values. A continuous one, such as a height, a mass or a waiting time, can take any value in a range. You cannot list a probability for each value, because there are infinitely many of them. Something has to replace the list.

The bags of rice show what. Group the 5000 masses into bars one gram wide, and the probability of landing in a group is the *area* of its bar. Now halve the bar width. There are twice as many bars, each about half as tall, and the total area is unchanged at 1. Halve it again, and again. The tops of the bars settle into a smooth curve, and the area under that curve is still 1.

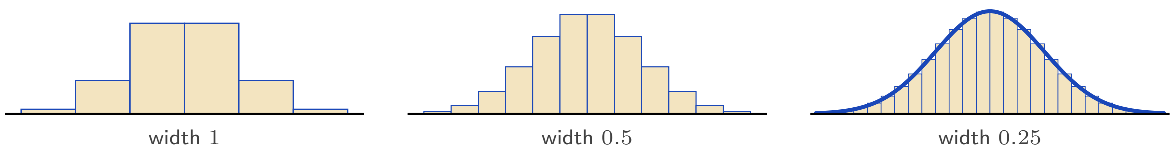

Figure 13.3 The same 5000 masses grouped into bars of decreasing width. Every version encloses total area 1; as the bars narrow, their tops approach a smooth curve. That curve is the density.

So probability for a continuous variable is **area**, and the curve whose area gives it is called a **probability density function**, written $f(x)$. Two conditions make f a valid density: it is never negative, and the total area under it is 1.

DEF A **probability density function** $f(x)$ satisfies

$$f(x) \geq 0 \quad \text{for all } x, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1,$$

and probabilities are areas: $P(a \leq X \leq b) = \displaystyle\int_a^b f(x)\,dx.$

One consequence looks strange at first. A bar of zero width has zero area, so $P(X = c) = 0$ for every single value c. This does not mean that value is impossible. It means no exact value carries a share of the total: a person's height is never exactly 170 cm to unlimited precision, only somewhere in a range near it. Since the endpoints contribute nothing, $P(a \leq X \leq b)$ and $P(a < X < b)$ are the same number, and you may be careless about $<$ and $\leq$ here in a way you cannot be with a discrete variable.

The expected value and variance carry over from the discrete case, with integrals replacing sums.

KEY

$$E(X) = \int_{-\infty}^{\infty} x\,f(x)\,dx, \qquad \text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

Three numbers describe where a density sits, and each is read off the curve in its own way. The **mode** is the value where f is highest. The **median** is the value that cuts the area in half. The **mean** is the balance point, $\int x f(x)\,dx$. For a symmetric curve all three coincide; for a lopsided one they do not, and the next example shows all three at work.

EXAMPLE 13.9

A continuous random variable has density $f(x) = kx$ for $0 \leq x \leq 2$, and $f(x) = 0$ elsewhere. Find k, then $P(X < 1)$, the mean, and the median.

Total area must be 1:

$$\int_0^2 kx\,dx = k\left[\tfrac{x^2}{2}\right]_0^2 = 2k = 1 \implies k = \tfrac{1}{2}.$$

Then

$$P(X<1)=\int_0^1 \tfrac{x}{2}\,dx=\left[\tfrac{x^2}{4}\right]_0^1=\tfrac{1}{4}.$$

The mean:

$$E(X)=\int_0^2 x\cdot\tfrac{x}{2}\,dx=\int_0^2 \tfrac{x^2}{2}\,dx=\left[\tfrac{x^3}{6}\right]_0^2=\tfrac{8}{6}=\tfrac{4}{3}.$$

The **median** m splits the area in half:

$$\int_0^m \tfrac{x}{2}\,dx=\tfrac{1}{2}\implies\tfrac{m^2}{4}=\tfrac{1}{2}\implies m=\sqrt{2}.$$

The **mode** is where f is greatest. Since $f(x) = \frac{x}{2}$ increases across $[0,2]$, the mode is at the right end, $x=2$.

CAUTION The mean, median, and mode of a continuous variable usually differ. The mode maximises $f(x)$; the median halves the area; the mean is the balance point $\int xf(x)\,dx$. Only for a symmetric density do all three coincide.

Piecewise densities

A density need not be a single formula. Any function that is non-negative and encloses total area 1 qualifies, including one built from pieces. The only new requirement is care with the arithmetic: integrals must be split at the joins, and the median may lie in either piece.

EXAMPLE 13.10

A continuous random variable has density

$$f(x)=\begin{cases}\dfrac{2x}{3} & 0\le x\le 1,\\ \dfrac{3-x}{3} & 1<x\le 3,\\ 0 & \text{otherwise.}\end{cases}$$

Verify that f is a valid density, find $P(X>2)$, and find the median.

Both pieces are non-negative, and the total area splits at $x=1$:

$$\int_0^1 \tfrac{2x}{3}\,dx+\int_1^3 \tfrac{3-x}{3}\,dx=\left[\tfrac{x^2}{3}\right]_0^1+\tfrac{1}{3}\left[3x-\tfrac{x^2}{2}\right]_1^3=\tfrac{1}{3}+\tfrac{2}{3}=1.\checkmark$$

For $P(X>2)$ only the second piece matters:

$$P(X>2)=\int_2^3 \tfrac{3-x}{3}\,dx=\tfrac{1}{3}\left[3x-\tfrac{x^2}{2}\right]_2^3=\tfrac{1}{6}.$$

For the median, first locate it: the area up to $x=1$ is $\frac{1}{3}<\frac{1}{2}$, so the median lies in the second piece. Collect the remaining area there:

$$\tfrac{1}{3} + \int_1^m \tfrac{3-x}{3}\,dx = \tfrac{1}{2} \implies 3m - \tfrac{m^2}{2} - \tfrac{5}{2} = \tfrac{1}{2} \implies m^2 - 6m + 6 = 0,$$

giving $m = 3 - \sqrt{3} \approx 1.27$ (taking the root inside the interval). The mode is $x = 1$, where the two pieces meet at the peak of the triangle.

Modes by calculus

When a density rises and then falls, its mode is an interior maximum, and you find it by the standard method: differentiate and set $f'(x) = 0$. The mode can also sit at an endpoint or at a corner, as in the examples above, so always check where the maximum actually is.

EXAMPLE 13.11

A continuous random variable has density $f(x) = kx^2(3-x)$ for $0 \le x \le 3$, and zero elsewhere. Find k, the mode, and $E(X)$.

Total area 1:

$$\int_0^3 k(3x^2 - x^3)\,dx = k\Big[x^3 - \tfrac{x^4}{4}\Big]_0^3 = k\left(27 - \tfrac{81}{4}\right) = \tfrac{27k}{4} = 1 \implies k = \tfrac{4}{27}.$$

For the mode, maximise f:

$$f'(x) = k(6x - 3x^2) = 3kx(2-x) = 0 \implies x = 0 \text{ or } x = 2.$$

Since f rises on $(0,2)$ and falls on $(2,3)$, the mode is $x = 2$. Finally,

$$E(X) = \int_0^3 k(3x^3 - x^4)\,dx = \tfrac{4}{27}\Big[\tfrac{3x^4}{4} - \tfrac{x^5}{5}\Big]_0^3 = \tfrac{4}{27} \cdot \tfrac{243}{20} = \tfrac{9}{5}.$$

Mean 1.8, mode 2: the density is not symmetric, so the balance point sits below the peak.

YOUR TURN 13.4 Continuous Random Variables HL

9.

(a) A density is $f(x) = kx^2$ for $0 \le x \le 3$, and zero elsewhere. Find k. [2]

(b) A density is $f(x) = k$ for $0 \le x \le 4$, and zero elsewhere. Find k and $P(X < 1)$. [2]

(c) A density is $f(x) = \dfrac{x}{8}$ for $0 \le x \le 4$, and zero elsewhere. Find $P(X > 2)$. [2]

(d) For the density in part (c), find $E(X)$. [3]

(e) A density is $f(x) = 3x^2$ for $0 \le x \le 1$, and zero elsewhere. Find the median. [3]

10.

(a) A continuous random variable has density $f(x) = kx(4-x)$ for $0 \le x \le 4$, and zero elsewhere. Find k. [3]

(b) Write down the mode of this density, and justify your answer. [2]

(c) Find $E(X)$. [2]

(d) A second variable has density $f(x) = x$ for $0 \le x \le 1$, $f(x) = \frac{1}{2}$ for $1 < x \le 2$, and zero elsewhere. Verify that the total area is 1, and find $P(X < 1.5)$. [4]

(e) Find the median of the density in part (d). [2]

13.5 The Normal Distribution

Refine the histogram of bag masses far enough and the bars settle into one particular density. It is the **normal distribution**, and everything from the last section applies to it: probability is area, and the total area is 1. What makes it worth a section of its own is how often it turns up — heights, masses, measurement errors, examination marks — so much of the natural and measured world that it appears in almost every area of statistics.

Two numbers fix a normal distribution completely, and it is worth being clear about what each one does.

- The **mean** μ locates the centre. Changing μ slides the whole curve left or right without altering its shape at all.
- The **standard deviation** σ sets the width. Increasing σ stretches the curve sideways, and because the area underneath must stay equal to 1, stretching it sideways also flattens it. A large σ therefore gives a wide, low bell and a small σ a narrow, tall one.

We write $X \sim N(\mu, \sigma^2)$, with the *variance* in the second slot, so remember to take a square root before using σ. The curve is symmetric about μ, so the mean, median and mode all sit together at the peak.

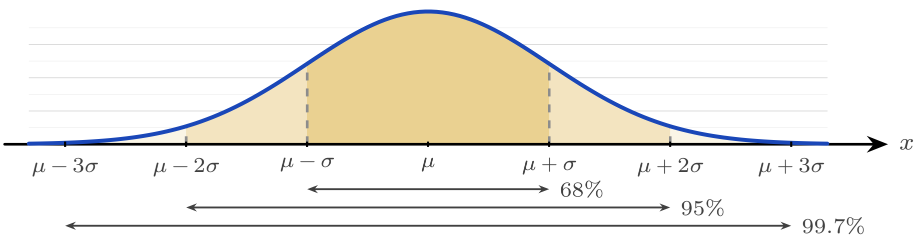

Figure 13.4 The normal curve, symmetric about μ. The three shaded bands reach one, two and three standard deviations either side of the mean, and hold about 68%, 95% and 99.7% of the values. Almost nothing lies beyond three standard deviations.

This symmetry gives a quick approximation, the **68–95–99.7 rule**: about 68% of values lie within one standard deviation of the mean, about 95% within two, and about 99.7% within three. It lets you judge, from μ and σ alone, where most of the data lie, without a calculator.

Read from the other end, the same rule measures how surprising a value is. Falling outside two standard deviations happens to about 5% of values, roughly one in twenty. Falling outside three happens to about 0.3%, roughly one in 370. That is why a reading three standard deviations from the mean is treated as a signal that something has changed, rather than as ordinary variation.

The curve has an equation, and although you will not have to use it, seeing it explains where the shape comes from.

KEY **The normal density** (not assessed):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Read the exponent first, because it carries the whole shape. The quantity $(x-\mu)^2$ is a parabola: it is zero at $x = \mu$ and grows as x moves away from the mean, and it depends only on the *distance* from μ, not on the direction. The minus sign turns that growth into decay, since e^{-u} is largest when $u = 0$ and shrinks towards zero as u grows. So the curve is tallest at the mean, falls away at the same rate on both sides, and never quite reaches the axis. The $2\sigma^2$ underneath sets the speed of the fall: a large σ makes the exponent grow slowly, giving a wide, low bell. The factor $\dfrac{1}{\sigma\sqrt{2\pi}}$ in front changes no part of the shape; it is the number that makes the total area exactly 1.

This formula is not in the formula booklet, and no examination question will ask you to use it. Every normal probability in this course is found with a GDC. It is here so that the bell is a consequence of something rather than a shape you are asked to accept.

For exact probabilities you use a GDC, which computes the area under the curve between any two values directly from μ and σ.

EXAMPLE 13.12

The heights of adult women are modelled by $X \sim N(170, 8^2)$ (in cm). Find (a) $P(X < 180)$, (b) $P(160 < X < 180)$.

Both are areas under the normal curve, read from a GDC with $\mu = 170$, $\sigma = 8$.

(a) $P(X < 180) \approx 0.894$.

(b) $P(160 < X < 180) \approx 0.789$.

The second answer is close to the 68% from the rule of thumb, because 160 and 180 are roughly one standard deviation either side of the mean.

YOUR TURN 13.5 The Normal Distribution

11. GDC

(a) $X \sim N(50, 10^2)$. Find $P(X < 60)$. [2]
(b) $X \sim N(100, 15^2)$. Find $P(X > 120)$. [2]
(c) $X \sim N(20, 4^2)$. Find $P(16 < X < 24)$. [2]
(d) $Z \sim N(0, 1)$. Find $P(Z < 1.5)$. [1]
(e) $X \sim N(500, 50^2)$. Find $P(X < 400)$. [2]

12. GDC The masses of packets of rice are modelled by $X \sim N(500, 25^2)$, in grams.

(a) Find the probability that a packet weighs more than 540 g. [2]
(b) Find $P(475 < X < 525)$, and compare your answer with the 68% from the rule of thumb. [3]
(c) In a batch of 800 packets, find the expected number weighing more than 540 g. [2]
(d) Write down the range of masses within which about 95% of packets lie. [2]

13.6 Standardization and the Inverse Normal

Every normal distribution is the same bell, just shifted and scaled. **Standardizing** expresses any value as the number of standard deviations it sits from its mean. That number is the **z-value**, and it converts any normal variable to the **standard normal** $Z \sim N(0, 1)$.

KEY · IB

$$z = \frac{x - \mu}{\sigma}$$

A z-value of 1.5 means "one and a half standard deviations above the mean," regardless of the original units. Standardizing lets you compare values from different normal distributions, and it is the key to working backwards.

EXAMPLE 13.13

Exam scores are $N(60, 12^2)$. A student scores 78. How many standard deviations above the mean is this?

$$z = \frac{78 - 60}{12} = \frac{18}{12} = 1.5.$$

The score sits 1.5 standard deviations above the mean. A student scoring 78 on a different exam with $N(70, 4^2)$ would have $z = 2$, and so be higher *relative to their cohort*, despite the equal raw mark.

The inverse normal

Often the question runs the other way: you know a probability and want the value that produces it. "What score is the 90th percentile?" The **inverse normal** on a GDC takes a probability and returns the corresponding value x.

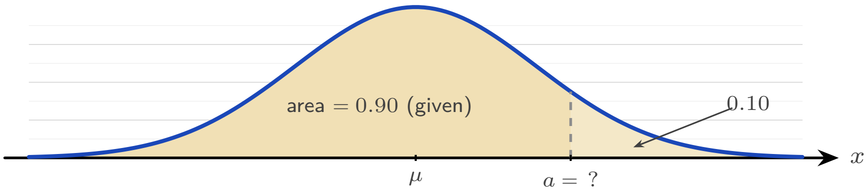

Figure 13.5 The inverse normal reverses the usual question. Instead of being given a and finding the area, you are given the area and must find a. A calculator expects the area to the *left* of a, so a question phrased as a right-hand tail, here 0.10, must be turned into its left-hand partner 0.90 first.

EXAMPLE 13.14

For heights $X \sim N(170, 8^2)$, find the height a below which 90% of women fall.
You need a with $P(X < a) = 0.90$. The inverse normal gives

$$a \approx 180.3 \text{ cm}.$$

Equivalently, the 90th percentile sits at $z \approx 1.282$, so $a = 170 + 1.282(8) \approx 180.3$, matching.

Finding an unknown μ or σ

The inverse normal also lets you recover a missing parameter. If you know a probability and a value, standardize to get a z, then solve the equation $z = \frac{x-\mu}{\sigma}$ for whichever of μ or σ is unknown.

EXAMPLE 13.15

A machine fills bottles so that volumes are normal with $\sigma = 5$ mL. Given that 80% of bottles contain less than 20 mL, find the mean μ.
Here $P(X < 20) = 0.80$. The inverse normal for the standard curve gives the matching z:

$$z = 0.8416, \qquad \text{so} \qquad 0.8416 = \frac{20-\mu}{5}.$$

Solving, $20 - \mu = 4.21$, hence $\mu \approx 15.8$ mL.

EXAMPLE 13.16

Scores are normal with mean 50. Given that 10% of scores exceed 60, find σ.

$P(X > 60) = 0.10$ means $P(X < 60) = 0.90$, so $z = 1.282$:

$$1.282 = \frac{60-50}{\sigma} \implies \sigma = \frac{10}{1.282} \approx 7.80.$$

Both parameters unknown

When neither μ nor σ is known, one probability statement is not enough: you need two. Each statement standardizes to an equation of the form $x = \mu + z\sigma$, and the pair solves simultaneously, exactly like any two linear equations.

EXAMPLE 13.17

The masses of a species of fish are normally distributed. It is known that 5% of the fish have mass below 50 g and 10% have mass above 80 g. Find μ and σ.

Translate each statement into a z-value with the inverse normal:

$$P(X < 50) = 0.05 \implies z_1 = -1.645, \qquad P(X < 80) = 0.90 \implies z_2 = 1.282.$$

Each gives an equation $x = \mu + z\sigma$:

$$50 = \mu - 1.645\sigma, \qquad 80 = \mu + 1.282\sigma.$$

Subtracting, $30 = 2.926\sigma$, so $\sigma \approx 10.3$ g. Substituting back, $\mu = 50 + 1.645(10.25) \approx 66.9$ g.

Note the signs carefully: a value *below* the mean has a negative z. One wrong sign here changes both answers, so check each z against the picture before solving.

YOUR TURN 13.6 Standardization and the Inverse Normal

13. GDC

(a) $X \sim N(60, 12^2)$. Find the z-value of $x = 84$. [2]

(b) $X \sim N(70, 5^2)$. Find the value x for which $P(X < x) = 0.95$. [2]

(c) $X \sim N(80, 10^2)$. Find the value x for which $P(X > x) = 0.15$. [3]

14. GDC

(a) $X \sim N(\mu, 8^2)$ with $P(X < 50) = 0.30$. Find μ. [3]

(b) $X \sim N(\mu, 6^2)$ with $P(X > 40) = 0.20$. Find μ. [3]

(c) $X \sim N(100, \sigma^2)$ with $P(X < 90) = 0.25$. Find σ. [3]

15. GDC

(a) Two examinations have marks $A \sim N(65, 10^2)$ and $B \sim N(72, 6^2)$. A student scores 75 in A and 80 in B. Find both z-values, and state which is the better performance relative to the cohort. [4]

(b) $X \sim N(\mu, \sigma^2)$ with $P(X < 20) = 0.10$ and $P(X < 35) = 0.90$. Find μ and σ. [5]

(c) For $X \sim N(50, 8^2)$, find the lower quartile, the upper quartile, and the interquartile range. [3]

Chapter 13 · Probability Distributions

Reflections

RW The bell curve's universality has a name: the central limit theorem. It says that the sum of many independent random effects tends toward a normal distribution, whatever the shape of the individual effects. This is why measurement errors, heights, and the means of large samples are all approximately normal. The pattern was noticed in 1846; the theorem that explains it was not fully proven until the 1930s.

TOK In 1846 a Belgian astronomer found this curve in the chest measurements of 5738 soldiers, and drew from it a conclusion the data did not support. He invented *l'homme moyen*, the "average man," and treated this statistical fiction as an ideal that real people deviate from. No actual person has all the average values at once. When we summarise a population by its mean, are we describing a real thing, or creating one? The normal curve is a fact about variation; the "average man" was an interpretation added to it.

TOK A game is called fair when the expected gain is zero. Every casino game fails that test by design, and the house edge is published.

If both sides know the odds, is a game with a built-in edge unfair, or merely unequal? And does the answer change when the player cannot do the arithmetic?

TOK This chapter offers several models for the same kind of situation: a binomial count can be approximated by a normal curve, and a real data set is never exactly either.

What criteria should decide between two models that both fit? Simplicity, accuracy, and agreement with existing theory can point different ways, and the choice is made by people.

EXT The binomial becoming normal is a theorem, not a loose comparison. Plot $B(n, 0.5)$ for increasing n and the discrete bars take the shape of the continuous bell. Galton built a machine, the quincunx, to show it physically: balls dropped through rows of pegs, each bounce an independent left-or-right trial, pile up at the bottom in a binomial that looks exactly normal. The distributions of this chapter are not separate facts but one story told at increasing scale.

End-of-Chapter Problems — Chapter 13: Probability Distributions

1. A discrete random variable X has the distribution below.

x	1	2	3	4
$P(X = x)$	0.1	k	0.3	0.2

(a) Find k. [2]

(b) Find $E(X)$. [2]

(c) Find Var(X). [3]

2. A charity raffle sells 500 tickets at \$2 each. There is one prize of \$300 and five prizes of \$50.

(a) Find the expected winnings (before cost) for a single ticket. [3]

(b) Hence find the expected gain or loss per ticket, and comment. [2]

3. GDC In a production line, 8% of items are faulty, independently. A random sample of 15 items is taken. Let X be the number of faulty items.

(a) State the distribution of X. [1]

(b) Find $P(X = 0)$. [2]

(c) Find $P(X \geq 2)$. [2]

(d) Find the expected number of faulty items. [1]

4. GDC A fair six-sided die is rolled 30 times. Let X be the number of sixes.

(a) State the distribution of X and find its mean. [2]

(b) Find $P(X = 5)$. [2]

(c) Find $P(X \geq 7)$. [2]

5. A continuous random variable has density $f(x) = kx$ for $0 \leq x \leq 3$, and zero elsewhere.

(a) Find k. [2]

(b) Find $P(X < 2)$. [2]

(c) Find $E(X)$. [3]

(d) Find the median and the mode. [3]

6. GDC The heights of a species of plant are $X \sim N(170, 8^2)$ in cm.

(a) Find $P(X < 160)$. [2]

(b) Find $P(160 < X < 185)$. [2]

(c) Find the proportion of plants taller than 180 cm. [2]

7. GDC Test scores are normally distributed with mean 500 and standard deviation 80.

(a) Find the score at the 90th percentile. [2]

(b) The top 5% of candidates receive a distinction. Find the minimum score for a distinction. [3]

8. GDC A variable X is normally distributed. It is known that $P(X < 40) = 0.10$ and $P(X < 70) = 0.75$.

(a) Write down two equations involving μ and σ using standardized values. [3]

(b) Find μ and σ. [4]

9. The random variable X has $E(X) = 5$ and $\text{Var}(X) = 4$. Let $Y = 2X - 1$.

(a) Find $E(Y)$. [1]

(b) Find $\text{Var}(Y)$ and the standard deviation of Y. [3]

10. GDC Reaction times are $X \sim N(60, 12^2)$ in milliseconds. A reaction is "slow" if it exceeds 75 ms.

(a) Find the probability that a single reaction is slow. [2]

(b) Thirty reactions are recorded independently. Find the expected number of slow reactions. [2]

(c) Find the probability that at least one of the thirty is slow. [3]

11. A continuous random variable has density $f(x) = \frac{1}{8}(4 - x)$ for $0 \leq x \leq 4$, and zero elsewhere.

(a) Verify that f is a valid probability density function. [3]

(b) Find $P(X < 2)$. [2]

(c) Find the mode. [1]

12. A discrete random variable X takes the values $1, 2, 3, 4$ with $P(X = 1) = 0.2$, $P(X = 2) = a$, $P(X = 3) = b$, and $P(X = 4) = 0.1$. It is known that $E(X) = 2.4$.

(a) Find the values of a and b. [4]

(b) Find $P(X > 2 \mid X \geq 2)$. [3]

13. A stall game costs $\$c$ to play. A player wins \$10 with probability 0.2, wins \$2 with probability 0.3, and otherwise wins nothing.

(a) Find the expected winnings from a single play. [2]
(b) Find the value of c for which the game is fair. [1]
(c) The stall sets $c = 3$. Find the player's expected loss over 100 plays. [2]

14. GDC A gardener plants 12 seeds. Each seed germinates with probability 0.85, independently of the others. Let X be the number that germinate.

(a) State two conditions that make a binomial model appropriate here. [2]
(b) Find $P(X = 10)$. [2]
(c) Find $P(X \geq 10)$. [2]
(d) Find the mean and standard deviation of X. [2]

15. GDC A darts player hits the bullseye with probability 0.3 on each throw, independently.

(a) In 8 throws, find the probability of exactly 3 bullseyes. [2]
(b) Find the least number of throws needed for the probability of at least one bullseye to exceed 0.98. [3]

16. GDC The random variable $X \sim B(10, p)$ satisfies $P(X = 0) = 0.0282$.

(a) Find the value of p. [3]
(b) Find $P(X = 3)$. [2]
(c) Write down $E(X)$ and Var(X). [2]

17. GDC The lifetime of a battery is $X \sim N(120, 15^2)$, in hours.

(a) Find the probability that a battery lasts more than 140 hours. [2]
(b) Find $P(100 < X < 130)$. [2]
(c) A company ships 10 000 batteries. Find the expected number that last more than 140 hours. [2]

18. GDC The masses of apples are $X \sim N(160, 20^2)$, in grams. The heaviest 8% are sold as premium, and the lightest 5% are rejected.

(a) Find the minimum mass of a premium apple. [2]
(b) Find the maximum mass of a rejected apple. [2]
(c) Write down the proportion of apples that are neither premium nor rejected. [1]

19. GDC Journey times are normally distributed with mean 25 minutes and standard deviation σ. It is known that $P(X < 30) = 0.85$.

(a) Find σ. [3]
(b) Hence write down $P(X < 20)$, justifying your answer. [2]

20. GDC The masses of loaves of bread are normally distributed with unknown mean μ and standard deviation σ. It is known that 2% of loaves have mass below 300 g and 5% have mass above 340 g.

(a) Write down two equations for μ and σ using standardized values. [3]
(b) Find μ and σ. [3]

21. A continuous random variable has density

$$f(x) = \begin{cases} k & 0 \le x \le 1, \\ k(2-x) & 1 < x \le 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that $k = \frac{2}{3}$. [3]
(b) Sketch the graph of f. [2]
(c) Find $P(X > 1.5)$. [2]
(d) Find the median of X. [2]
(e) Find $E(X)$. [3]

22. A continuous random variable has density $f(x) = kx(1-x)$ for $0 \le x \le 1$, and zero elsewhere.

(a) Show that $k = 6$. [2]
(b) Use calculus to find the mode of X. [2]
(c) Find $E(X)$. [2]
(d) Find Var(X). [3]

23. GDC Scores on an examination are $X \sim N(55, 10^2)$. The pass mark is 45.

(a) Find the probability that a randomly chosen candidate passes. [2]
(b) Six candidates are chosen at random. Find the probability that exactly five of them pass. [3]
(c) Find the probability that at least five of them pass. [2]

24. GDC In 1846 a Belgian astronomer published the chest measurements, in inches, of 5738 Scottish militiamen, and used them as evidence that human measurements follow a normal distribution. His table is reproduced below.

Chest (in)	33	34	35	36	37	38	39	40
Frequency	3	18	81	185	420	749	1073	1079
Chest (in)	41	42	43	44	45	46	47	48
Frequency	934	658	370	92	50	21	4	1

(a) Show that the mean chest measurement is 39.8 inches, to 3 significant figures. [3]

(b) The standard deviation of the measurements is 2.05 inches. Assuming $X \sim N(39.8,\, 2.05^2)$, find $P(X < 38.5)$. [2]

(c) Find the proportion of the 5738 soldiers who actually measured 38 inches or less, and compare it with your answer to part (b). [3]

(d) Find the proportion of soldiers whose measurement lies within one standard deviation of the mean, and compare it with the value predicted by the normal model. [3]

(e) Comment on how well the normal distribution describes this real data set. [2]

Challenge Questions

25. What makes a game fair? At a fair, a stall offers this game. You pay a stake, roll two fair dice, and are paid $20 if the total is 12, $5 if the total is 10 or 11, and nothing otherwise. Let W be the amount paid out.

(a) Write down the probability distribution of W. [3]

(b) Find $E(W)$. [3]

(c) State the stake that would make the game fair, and explain what "fair" means here. [3]

(d) The stall charges $2. Find the expected profit per game for the stall, and the expected profit over 500 games. [3]

(e) Find Var(W) and the standard deviation of W. [3]

(f) A player who plays ten times often finishes ahead, even though the game favours the stall. Use your answer to part (e) to explain why, and explain why the stall is not worried. [4]

26. GDC **From the binomial to the bell.** The Reflections claim that a binomial distribution comes to look normal as n grows. This question tests that claim, and finds the adjustment that makes it work.

Throughout, let $X \sim B(50, 0.5)$ and let $Y \sim N(\mu, \sigma^2)$ have the same mean and variance as X.

(a) Write down $E(X)$ and Var(X), and hence state μ and σ for Y. [2]

(b) Find $P(20 \leq X \leq 30)$ using the binomial. [2]

(c) Find $P(20 < Y < 30)$ using the normal. Comment on how well it matches part (b). [3]

(d) X takes whole-number values, while Y takes every value in between. The bar of the histogram for $X = 20$ covers the interval from 19.5 to 20.5. Using this idea, find $P(19.5 < Y < 30.5)$ and compare it with part (b). [4]

(e) Repeat parts (b), (c) and (d) for $X \sim B(10, 0.5)$ and the range $4 \leq X \leq 6$. [4]

(f) State which of the two normal calculations should be used, and describe how the quality of the approximation changes as n increases. [3]

27. The most likely value. Let $X \sim B(n, p)$ with $0 < p < 1$. This question finds the mode of X, the value it takes most often.

(a) Show that for $r = 1, 2, \dots, n$,

$$\frac{P(X = r)}{P(X = r - 1)} = \frac{(n - r + 1)\,p}{r\,(1 - p)}.$$

[3]

(b) Hence show that $P(X = r) \geq P(X = r - 1)$ if and only if $r \leq (n + 1)p$. [3]

(c) Deduce that the mode of X is $\lfloor (n + 1)p \rfloor$ when $(n + 1)p$ is not an integer, and use this to find the most likely number of successes when $X \sim B(20, 0.3)$. [3]

(d) Explain what happens when $(n + 1)p$ is an integer, and illustrate your answer with $X \sim B(9, 0.5)$. [3]